



**CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE
PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	CHEMISTRY IN EVERYDAY LIFE				
Type of Course	SEC				
Semester	V				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	1. Fundamental Chemistry 1. Polymers- Natural and synthetic				
Course Summary	This course opens the the vast domain of applied Chemistry for all				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand the composition of products used in everyday life	U	F	Instructed created exams, quiz
CO2	To create awareness on the safety regulations of food products	An	C	Seminars
CO3	To gain knowledge on beverages	U	C	Assignment
CO4	To develop environmentally friendly polymers	An	P	Observation of practical skill
CO5	To apply eco-friendly plastic disposal methods	E	P	Exams
CO6	To demonstrate efficient energy storage systems	U	F	Assignment/ppt presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Diary Products and Beverages		10	20
	1	Composition of milk and milk products.	2	
	2	Analysis of fat content, minerals in milk and butter.	2	
	3	Estimation of added water in milk.	1	
	4	Beverages: Analysis of caffeine in coffee and tea,	2	
	5	Detection of chicory in coffee	1	
	6	Chloral hydrate in toddy,	1	
	7	Determination of methyl alcohol in alcoholic beverages	1	
II	Food additives		10	20
	8	Food additives – definitions, classification, and function	1	
	9	Antioxidants, Preservatives, Emulsifiers, Stabilizers, sweeteners, thickening agents, chelating agents, curing agents, leavening agents, anti-caking agents, colouring agents	2	
	10	Flavouring agents, stimulants. Functional rule of food additives	2	
	11	Safety and regulations of food additives.	2	
	12	Food allergy and intolerance	2	
	13	Benefits of additives- Side effects of food additives	1	
III	Polymers		10	20
	14	Basic concept of polymer- classification and characteristics of polymers	2	
	15	Applications of polymers as plastics in electronics, automobile components, medical fields and aerospace materials.	3	
	16	Problems of plastic waste management.	2	
	17	Strategies for the development of environment-friendly polymers	3	
IV	Chemical and Renewable Energy Sources:		6	10
	18	Principles and applications of primary & secondary batteries and fuel Cells	3	

	19	Basics of solar energy	3	
V		Open Ended: (Practical experience may be offered)	9	
		Analysis of milk, beverages Synthesis of a polymer A project on Food labels and the actual contents Review project on solar cells and batteries		

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	-	-	-	3	-	1	-	1	-
CO 2	2	3	-	-	-	-	2	-	-	-	-	2
CO 3	-	-	1	-	-	-	2	-	-	-	-	-
CO 4	-	-	2	3	-	-	2	-	2	-	-	2
CO 5	-	1	-	-	-	-	-	-	-	-	-	2
CO 6	-	-	-	3	-	-	2	-	1	-	-	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	

References

1. B. K. Sharma: Introduction to Industrial Chemistry, Goel Publishing, Meerut (1998)
2. Ashtoush Kar. Medicinal Chemistry (Two Colour Edition), New Age International Pvt Ltd, 2022
3. Edward Cox Henry , The Chemical analysis of Foods , Hardcover , Hassell Street Press , 2021
4. Fred Billmeyer : Textbook of polymer science; Wiley 3rd addition.